

# Conclusion to Regular Expressions

 <https://github.com/learn-co-curriculum/python-p3-conclusion-to-regex>  <https://github.com/learn-co-curriculum/python-p3-conclusion-to-regex/issues/new>

## Learning Goals

- Validate that strings match specific patterns using regular expressions.
- Search for strings that match specific patterns using regular expressions.

## Key Vocab

- **Regular Expression:** a sequence of characters used to search for a pattern inside of a string.
- **Pattern:** a description of sequences of characters that share certain traits with one another. Sequences do not need to be the same length or share any common characters to pattern match. Also called a **filter**.

## Conclusion

Regular expressions can be tricky, but there are many tools that will help you craft patterns to check that strings match important patterns (like email addresses and phone numbers) and search for these types of strings in massive walls of text. We learned that:

- There is a near-universal regular expression language that stems from PERL.
- There are online tools to check your regular expression patterns as you go, such as [regex101](https://regex101.com)  <https://regex101.com>.
- RegExes in Python are managed using the `re` module in the Python standard library.
- `search()`, `match()`, and `fullmatch()` return a single `re.Match` object if your RegEx pattern has a match in the search string.
- `findall()` returns a list of matching strings in the search string.
- `split()` and `sub()` can be used to modify strings where there are matches for a RegEx pattern.

We didn't cover everything about regular expressions in this module- there are many nuances that you will learn throughout your software career. Make sure to revisit regular expressions *regularly* to keep building your skills.

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## Resources

- [Python 3 Documentation](https://docs.python.org/3/) ➞ [\(https://docs.python.org/3/\)](https://docs.python.org/3/)
- [re - Regular expression operations - Python](https://docs.python.org/3/library/re.html) ➞ [\\_re.html\)](https://docs.python.org/3/library/re.html)
- [regex101](https://regex101.com) ➞ [\\_\]\(https://regex101.com\)\]](https://regex101.com)
- [Python Regular Expressions - Google for Education](https://developers.google.com/edu/python/regular-expressions) ➞ [\\_\]\(https://developers.google.com/edu/python/regular-expressions\)](https://developers.google.com/edu/python/regular-expressions)